

- 1 Background to the Project
- 2 Data Import and Initial Exploration
- 3 Exploratory Data Analysis
- 4 Correlation Analysis
- 5 Poor Performer Classification
- 7 Executive Summary

Project 3: What Characteristics Are Associated with Poor Hospital Performance?

Merrimack College DSE6630: Healthcare & Life Sciences Analytics

Peter Gyorda

26 June 2026

1 Background to the Project

1.1 Overview

Hospital quality has become an increasingly important focus for healthcare organizations, patients, and policymakers. The Centers for Medicare & Medicaid Services (CMS) publicly reports a variety of quality measures, including mortality, readmission, and patient safety outcomes. These measures provide an opportunity to identify characteristics associated with poor-performing hospitals and to better understand factors that may influence quality of care.

This project analyzes publicly available CMS hospital quality data to determine whether hospital characteristics, such as ownership, hospital type, emergency service availability, and geographic location, are associated with poor performance. The analysis also explores whether these characteristics can be used to predict poor-performing hospitals using machine learning techniques.

1.2 Background

CMS reports numerous quality measures for hospitals across the United States. Rather than relying solely on an overall hospital rating, this project creates a custom **Poor Performance Score** by combining the number of mortality, readmission, and patient safety measures in which each hospital performed

worse than the national average. This approach provides a more detailed assessment of hospital performance across multiple quality domains.

1.3 Project Objectives

The objectives of this project are to:

- Explore the distribution of hospital quality performance across the United States.
- Identify hospital characteristics associated with poor performance.
- Evaluate differences in performance based on hospital ownership, hospital type, and emergency service availability.
- Examine geographic variation in hospital quality.
- Develop a Random Forest model to determine whether publicly available hospital characteristics can predict poor-performing hospitals.
- Provide recommendations that hospital administrators can use to prioritize quality improvement efforts.

1.4 Stakeholder

The primary stakeholders for this analysis are hospital administrators and healthcare system leadership teams seeking to identify organizational characteristics associated with poor hospital performance and opportunities for quality improvement.

2 Data Import and Initial Exploration

2.1 Load the CMS Hospital Dataset

```
library(readr)
```

```
hospital <- read_csv("Hospital_General_Information.csv")
```

```
## Rows: 5394 Columns: 39
## — Column specification —————
## Delimiter: ","
## chr (34): Facility ID, Facility Name, Address, City/Town, State, ZIP Code, C...
## dbl (5): MORT Group Footnote, Safety Group Footnote, READM Group Footnote, ...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

2.2 Examine the Dataset

```
dim(hospital)
```

```
## [1] 5394 39
```

```
names(hospital)
```

```
## [1] "Facility ID"
## [2] "Facility Name"
## [3] "Address"
## [4] "City/Town"
## [5] "State"
## [6] "ZIP Code"
## [7] "County/Parish"
## [8] "Telephone Number"
## [9] "Hospital Type"
## [10] "Hospital Ownership"
## [11] "Emergency Services"
## [12] "Meets criteria for promoting interoperability of EHRs"
## [13] "Meets criteria for birthing friendly designation"
## [14] "Hospital overall rating"
## [15] "Hospital overall rating footnote"
## [16] "MORT Group Measure Count"
## [17] "Count of Facility MORT Measures"
## [18] "Count of MORT Measures Better"
## [19] "Count of MORT Measures No Different"
## [20] "Count of MORT Measures Worse"
## [21] "MORT Group Footnote"
## [22] "Safety Group Measure Count"
## [23] "Count of Facility Safety Measures"
## [24] "Count of Safety Measures Better"
## [25] "Count of Safety Measures No Different"
## [26] "Count of Safety Measures Worse"
## [27] "Safety Group Footnote"
## [28] "READM Group Measure Count"
## [29] "Count of Facility READM Measures"
## [30] "Count of READM Measures Better"
## [31] "Count of READM Measures No Different"
## [32] "Count of READM Measures Worse"
## [33] "READM Group Footnote"
## [34] "Pt Exp Group Measure Count"
## [35] "Count of Facility Pt Exp Measures"
## [36] "Pt Exp Group Footnote"
## [37] "TE Group Measure Count"
## [38] "Count of Facility TE Measures"
## [39] "TE Group Footnote"
```

```
glimpse(hospital)
```

```

## Rows: 5,394
## Columns: 39
## $ `Facility ID`                <chr> "010001", "010...
## $ `Facility Name`              <chr> "SOUTHEAST HEA...
## $ Address                      <chr> "1108 ROSS CLA...
## $ `City/Town`                  <chr> "DOTHAN", "BOA...
## $ State                       <chr> "AL", "AL", "A...
## $ `ZIP Code`                   <chr> "36301", "3595...
## $ `County/Parish`              <chr> "HOUSTON", "MA...
## $ `Telephone Number`           <chr> "(334) 793-870...
## $ `Hospital Type`              <chr> "Acute Care Ho...
## $ `Hospital Ownership`         <chr> "Government - ...
## $ `Emergency Services`         <chr> "Yes", "Yes", ...
## $ `Meets criteria for promoting interoperability of EHRs` <chr> "Y", "Y", "Y",...
## $ `Meets criteria for birthing friendly designation`   <chr> "Y", NA, "Y", ...
## $ `Hospital overall rating`    <chr> "3", "2", "1",...
## $ `Hospital overall rating footnote` <chr> NA, NA, NA, NA...
## $ `MORT Group Measure Count`   <chr> "7", "7", "7",...
## $ `Count of Facility MORT Measures` <chr> "7", "6", "7",...
## $ `Count of MORT Measures Better` <chr> "1", "0", "0",...
## $ `Count of MORT Measures No Different` <chr> "6", "5", "6",...
## $ `Count of MORT Measures Worse` <chr> "0", "1", "1",...
## $ `MORT Group Footnote`        <dbl> NA, NA, NA, NA...
## $ `Safety Group Measure Count` <chr> "8", "8", "8",...
## $ `Count of Facility Safety Measures` <chr> "7", "7", "7",...
## $ `Count of Safety Measures Better` <chr> "2", "0", "3",...
## $ `Count of Safety Measures No Different` <chr> "5", "7", "4",...
## $ `Count of Safety Measures Worse` <chr> "0", "0", "0",...
## $ `Safety Group Footnote`      <dbl> NA, NA, NA, NA...
## $ `READM Group Measure Count`  <chr> "11", "11", "1...
## $ `Count of Facility READM Measures` <chr> "11", "9", "9"...
## $ `Count of READM Measures Better` <chr> "1", "0", "0",...
## $ `Count of READM Measures No Different` <chr> "8", "8", "7",...
## $ `Count of READM Measures Worse` <chr> "2", "1", "2",...
## $ `READM Group Footnote`       <dbl> NA, NA, NA, NA...
## $ `Pt Exp Group Measure Count` <chr> "8", "8", "8",...
## $ `Count of Facility Pt Exp Measures` <chr> "8", "8", "8",...
## $ `Pt Exp Group Footnote`      <dbl> NA, NA, NA, NA...
## $ `TE Group Measure Count`     <chr> "12", "12", "1...
## $ `Count of Facility TE Measures` <chr> "10", "12", "1...
## $ `TE Group Footnote`         <dbl> NA, NA, NA, NA...

```

```
summary(hospital)
```

##	Facility ID	Facility Name	Address	City/Town
##	Length:5394	Length:5394	Length:5394	Length:5394
##	Class :character	Class :character	Class :character	Class :character
##	Mode :character	Mode :character	Mode :character	Mode :character
##				
##				
##				
##	State	ZIP Code	County/Parish	Telephone Number
##	Length:5394	Length:5394	Length:5394	Length:5394
##	Class :character	Class :character	Class :character	Class :character
##	Mode :character	Mode :character	Mode :character	Mode :character
##				
##				
##				
##	Hospital Type	Hospital Ownership	Emergency Services	
##	Length:5394	Length:5394	Length:5394	
##	Class :character	Class :character	Class :character	
##	Mode :character	Mode :character	Mode :character	
##				
##				
##				
##	Meets criteria for promoting interoperability of EHRs			
##	Length:5394			
##	Class :character			
##	Mode :character			
##				
##				
##				
##	Meets criteria for birthing friendly designation	Hospital overall rating		
##	Length:5394	Length:5394		
##	Class :character	Class :character		
##	Mode :character	Mode :character		
##				
##				
##				
##	Hospital overall rating footnote	MORT Group Measure Count		
##	Length:5394	Length:5394		
##	Class :character	Class :character		
##	Mode :character	Mode :character		
##				
##				
##				
##				

```

## Count of Facility MORT Measures Count of MORT Measures Better
## Length:5394 Length:5394
## Class :character Class :character
## Mode :character Mode :character
##
##
##
##
## Count of MORT Measures No Different Count of MORT Measures Worse
## Length:5394 Length:5394
## Class :character Class :character
## Mode :character Mode :character
##
##
##
##
## MORT Group Footnote Safety Group Measure Count
## Min. : 5.00 Length:5394
## 1st Qu.: 5.00 Class :character
## Median : 5.00 Mode :character
## Mean :10.94
## 3rd Qu.:19.00
## Max. :23.00
## NA's :3534
## Count of Facility Safety Measures Count of Safety Measures Better
## Length:5394 Length:5394
## Class :character Class :character
## Mode :character Mode :character
##
##
##
##
## Count of Safety Measures No Different Count of Safety Measures Worse
## Length:5394 Length:5394
## Class :character Class :character
## Mode :character Mode :character
##
##
##
##
## Safety Group Footnote READM Group Measure Count
## Min. : 5.0 Length:5394
## 1st Qu.: 5.0 Class :character
## Median : 5.0 Mode :character
## Mean :10.6
## 3rd Qu.:19.0
## Max. :23.0
## NA's :3413
## Count of Facility READM Measures Count of READM Measures Better

```

```

## Length:5394                      Length:5394
## Class :character                  Class :character
## Mode :character                   Mode :character
##
##
##
##
## Count of READM Measures No Different Count of READM Measures Worse
## Length:5394                      Length:5394
## Class :character                  Class :character
## Mode :character                   Mode :character
##
##
##
## READM Group Footnote Pt Exp Group Measure Count
## Min. : 5.0                      Length:5394
## 1st Qu.: 5.0                    Class :character
## Median :19.0                    Mode :character
## Mean :15.3
## 3rd Qu.:19.0
## Max. :22.0
## NA's :4326
## Count of Facility Pt Exp Measures Pt Exp Group Footnote TE Group Measure Count
## Length:5394                      Min. : 5.00          Length:5394
## Class :character                  1st Qu.: 5.00          Class :character
## Mode :character                   Median : 5.00          Mode :character
##                                  Mean :10.04
##                                  3rd Qu.:19.00
##                                  Max. :22.00
##                                  NA's :3212
## Count of Facility TE Measures TE Group Footnote
## Length:5394                      Min. : 5.00
## Class :character                  1st Qu.:19.00
## Mode :character                   Median :19.00
##                                  Mean :17.63
##                                  3rd Qu.:19.00
##                                  Max. :22.00
##                                  NA's :4523

```

2.3 Missing Data Assessment

```
colSums(is.na(hospital))
```

##	Facility ID	
##		0
##	Facility Name	
##		0
##	Address	
##		0
##	City/Town	
##		0
##	State	
##		0
##	ZIP Code	
##		0
##	County/Parish	
##		0
##	Telephone Number	
##		0
##	Hospital Type	
##		0
##	Hospital Ownership	
##		0
##	Emergency Services	
##		0
##	Meets criteria for promoting interoperability of EHRs	
##		1507
##	Meets criteria for birthing friendly designation	
##		3159
##	Hospital overall rating	
##		0
##	Hospital overall rating footnote	
##		2780
##	MORT Group Measure Count	
##		0
##	Count of Facility MORT Measures	
##		0
##	Count of MORT Measures Better	
##		0
##	Count of MORT Measures No Different	
##		0
##	Count of MORT Measures Worse	
##		0
##	MORT Group Footnote	
##		3534
##	Safety Group Measure Count	
##		0
##	Count of Facility Safety Measures	
##		0
##	Count of Safety Measures Better	
##		0

Verify the cleaned variables

```
summary(hospital_clean[, c("mort_worse",  
                           "readm_worse",  
                           "safety_worse")])
```

```
##      mort_worse      readm_worse      safety_worse  
## Min.       :0.0000   Min.       :0.0000   Min.       :0.0000  
## 1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.0000  
## Median :0.0000   Median :0.0000   Median :0.0000  
## Mean    :0.0998   Mean    :0.5201   Mean    :0.1463  
## 3rd Qu.:0.0000   3rd Qu.:1.0000   3rd Qu.:0.0000  
## Max.    :3.0000   Max.    :7.0000   Max.    :4.0000  
## NA's    :1857    NA's    :1068    NA's    :1976
```

The three variables were successfully converted to numeric values, with missing values representing hospitals where the measures were reported as “Not Available.”

2.5 Create the Poor Performance Score

```
hospital_clean <- hospital_clean %>%  
  mutate(  
    PoorPerformanceScore = rowSums(  
      cbind(  
        mort_worse,  
        readm_worse,  
        safety_worse  
      ),  
      na.rm = FALSE  
    )  
  )  
  
summary(hospital_clean$PoorPerformanceScore)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's  
## 0.0000  0.0000  1.0000  0.9747  1.2500  9.0000  2306
```

```
sum(is.na(hospital_clean$PoorPerformanceScore))
```

```
## [1] 2306
```

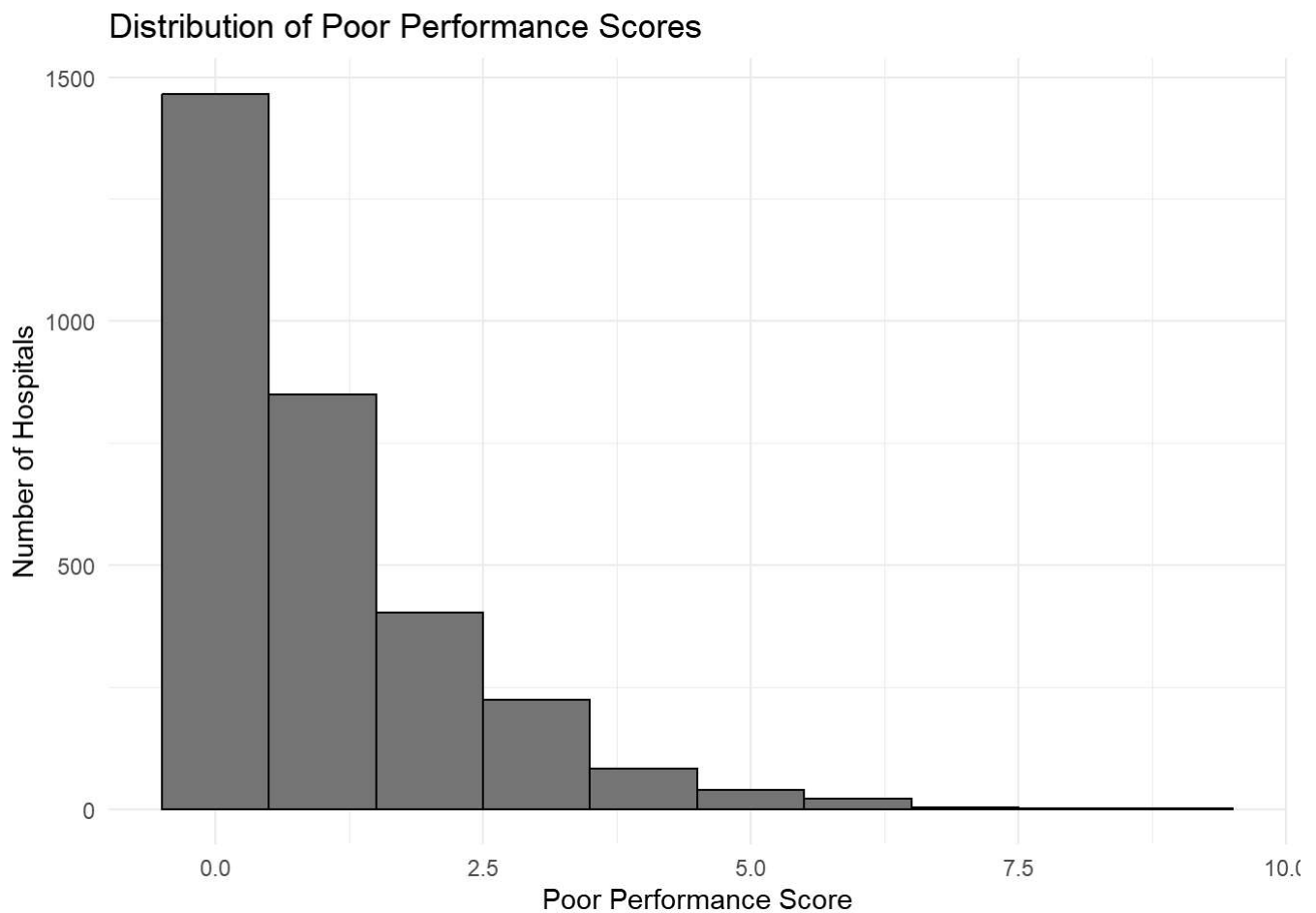
2.6 Remove Missing Observations

```
hospital_final <- hospital_clean %>%  
  filter(!is.na(PoorPerformanceScore))  
  
dim(hospital_final)
```

```
## [1] 3088  43
```

2.7 Distribution of the Poor Performance Score

```
ggplot(hospital_final,  
       aes(x = PoorPerformanceScore)) +  
  geom_histogram(binwidth = 1,  
                 fill = "steelblue",  
                 color = "black") +  
  labs(  
    title = "Distribution of Poor Performance Scores",  
    x = "Poor Performance Score",  
    y = "Number of Hospitals"  
  ) +  
  theme_minimal()
```



3 Exploratory Data Analysis

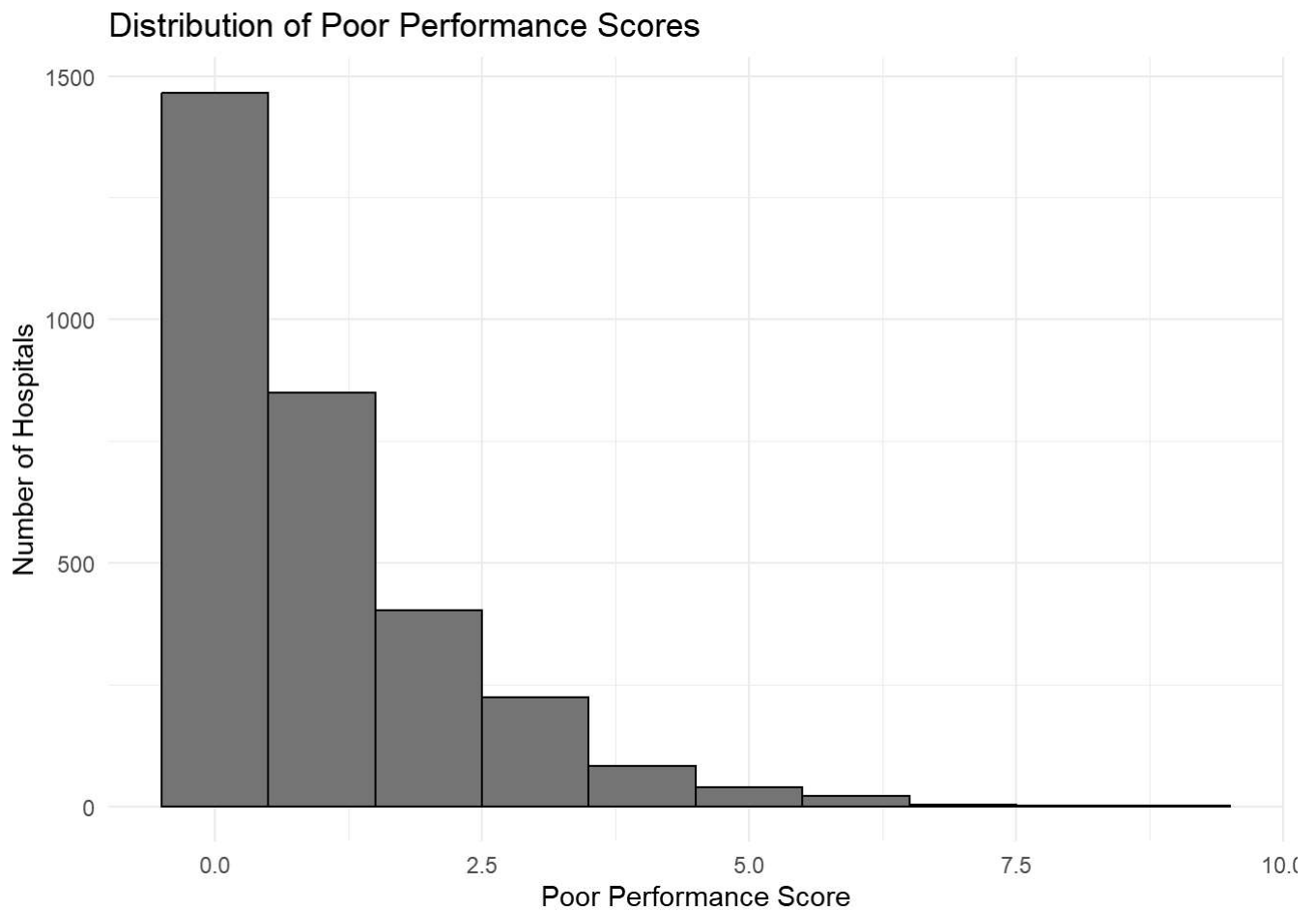
3.1 Distribution of the Poor Performance Score

```
# Remove hospitals with missing Poor Performance Scores
hospital_analysis <- hospital_clean %>%
  filter(!is.na(PoorPerformanceScore))
```

```
# Summary statistics
summary(hospital_analysis$PoorPerformanceScore)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.0000  0.0000  1.0000  0.9747  1.2500  9.0000
```

```
# Histogram
ggplot(hospital_analysis,
       aes(x = PoorPerformanceScore)) +
  geom_histogram(
    binwidth = 1,
    fill = "steelblue",
    color = "black"
  ) +
  labs(
    title = "Distribution of Poor Performance Scores",
    x = "Poor Performance Score",
    y = "Number of Hospitals"
  ) +
  theme_minimal()
```



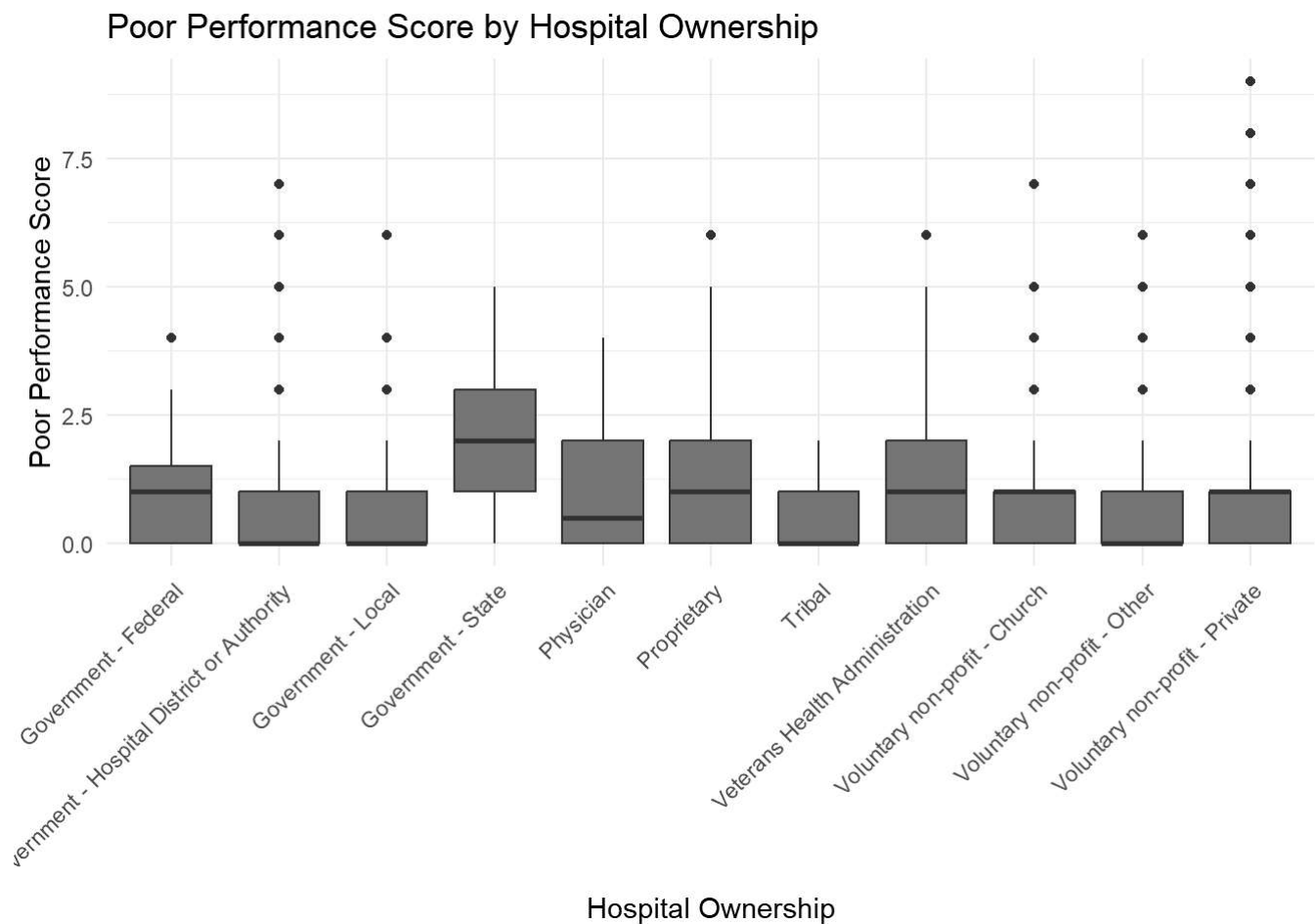
3.2 Hospital Ownership

```
ownership_summary <- hospital_analysis %>%
  group_by(`Hospital Ownership`) %>%
  summarise(
    Hospitals = n(),
    MeanScore = mean(PoorPerformanceScore),
    MedianScore = median(PoorPerformanceScore),
    SD = sd(PoorPerformanceScore)
  ) %>%
  arrange(desc(MeanScore))
```

ownership_summary

```
## # A tibble: 11 × 5
##   `Hospital Ownership` Hospitals MeanScore MedianScore SD
##   <chr>                <int>     <dbl>       <dbl> <dbl>
## 1 Government - State      35      2.09         2    1.62
## 2 Veterans Health Administration 109      1.24         1    1.35
## 3 Government - Federal    11      1.09         1    1.38
## 4 Physician              12      1.08         0.5  1.31
## 5 Proprietary            511      1.06         1    1.22
## 6 Voluntary non-profit - Private 1498      0.988        1    1.28
## 7 Voluntary non-profit - Church  235      0.928        1    1.15
## 8 Government - Hospital District or Auth... 234      0.919         0    1.36
## 9 Voluntary non-profit - Other    272      0.754         0    1.05
## 10 Government - Local    168      0.685         0    1.05
## 11 Tribal                 3      0.667         0    1.15
```

```
ggplot(hospital_analysis,
  aes(x = `Hospital Ownership`,
    y = PoorPerformanceScore)) +
  geom_boxplot(fill = "steelblue") +
  labs(
    title = "Poor Performance Score by Hospital Ownership",
    x = "Hospital Ownership",
    y = "Poor Performance Score"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(
      angle = 45,
      hjust = 1
    )
  )
```



Hospital Ownership. Government-owned state hospitals had the highest Poor Performance Scores. Most ownership groups had median scores of 0 or 1, although outliers were present across several ownership types.

3.3 Hospital Type

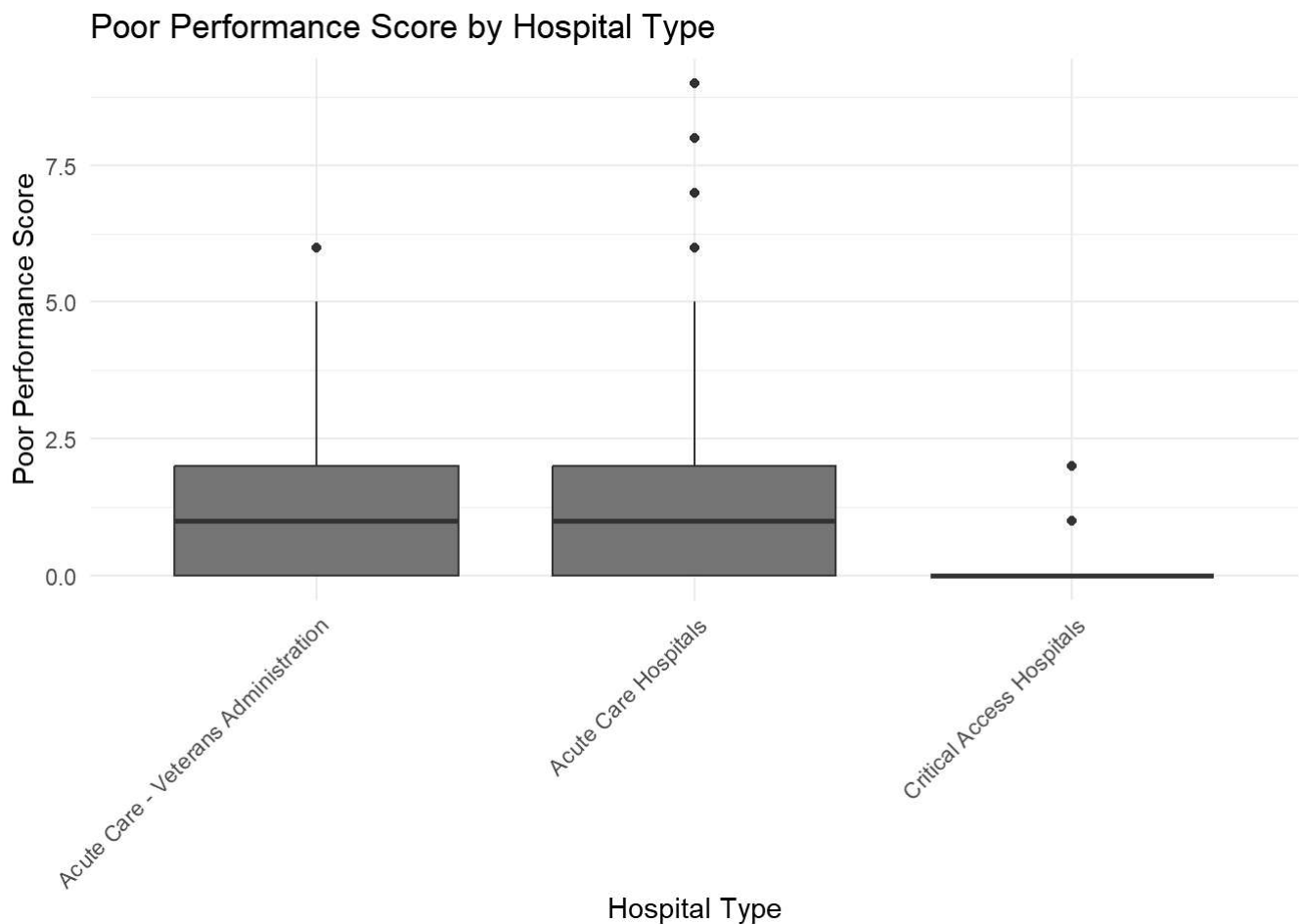
The Poor Performance Score was compared across hospital types to determine whether certain types of hospitals were associated with higher quality concerns.

```
type_summary <- hospital_analysis %>%
  group_by(`Hospital Type`) %>%
  summarise(
    Hospitals = n(),
    MeanScore = round(mean(PoorPerformanceScore), 2),
    MedianScore = round(median(PoorPerformanceScore), 2),
    SD = round(sd(PoorPerformanceScore), 2)
  ) %>%
  arrange(desc(MeanScore))
```

type_summary

```
## # A tibble: 3 × 5
##   `Hospital Type`      Hospitals MeanScore MedianScore    SD
##   <chr>              <int>      <dbl>      <dbl> <dbl>
## 1 Acute Care - Veterans Administration    109      1.24        1  1.35
## 2 Acute Care Hospitals                2676      1.06        1  1.28
## 3 Critical Access Hospitals              303      0.17        0  0.39
```

```
ggplot(hospital_analysis,
       aes(x = `Hospital Type`,
           y = PoorPerformanceScore)) +
  geom_boxplot(fill = "steelblue") +
  labs(
    title = "Poor Performance Score by Hospital Type",
    x = "Hospital Type",
    y = "Poor Performance Score"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(
      angle = 45,
      hjust = 1
    )
  )
)
```

Hospital Type. Hospital performance varied across hospital types. Differences in the median scores and several outliers suggest that hospital type may be associated with quality performance.

3.4 Emergency Services

The Poor Performance Score was compared between hospitals that provide emergency services and those that do not.

```
emergency_summary <- hospital_analysis %>%  
  group_by(`Emergency Services`) %>%  
  summarise(  
    Hospitals = n(),  
    MeanScore = round(mean(PoorPerformanceScore), 2),  
    MedianScore = round(median(PoorPerformanceScore), 2),  
    SD = round(sd(PoorPerformanceScore), 2)  
  )  
  
emergency_summary
```

```
## # A tibble: 2 × 5
##   `Emergency Services` Hospitals MeanScore MedianScore    SD
##   <chr>             <int>      <dbl>      <dbl> <dbl>
## 1 No                128        0.66          0  0.85
## 2 Yes              2960        0.99          1  1.27
```

Emergency Services. Hospitals with emergency services had higher Poor Performance Scores than hospitals without emergency services. The average score was 0.99 for hospitals with emergency services compared with 0.66 for hospitals without emergency services.

3.5 State Analysis

The average Poor Performance Score was calculated for each state.

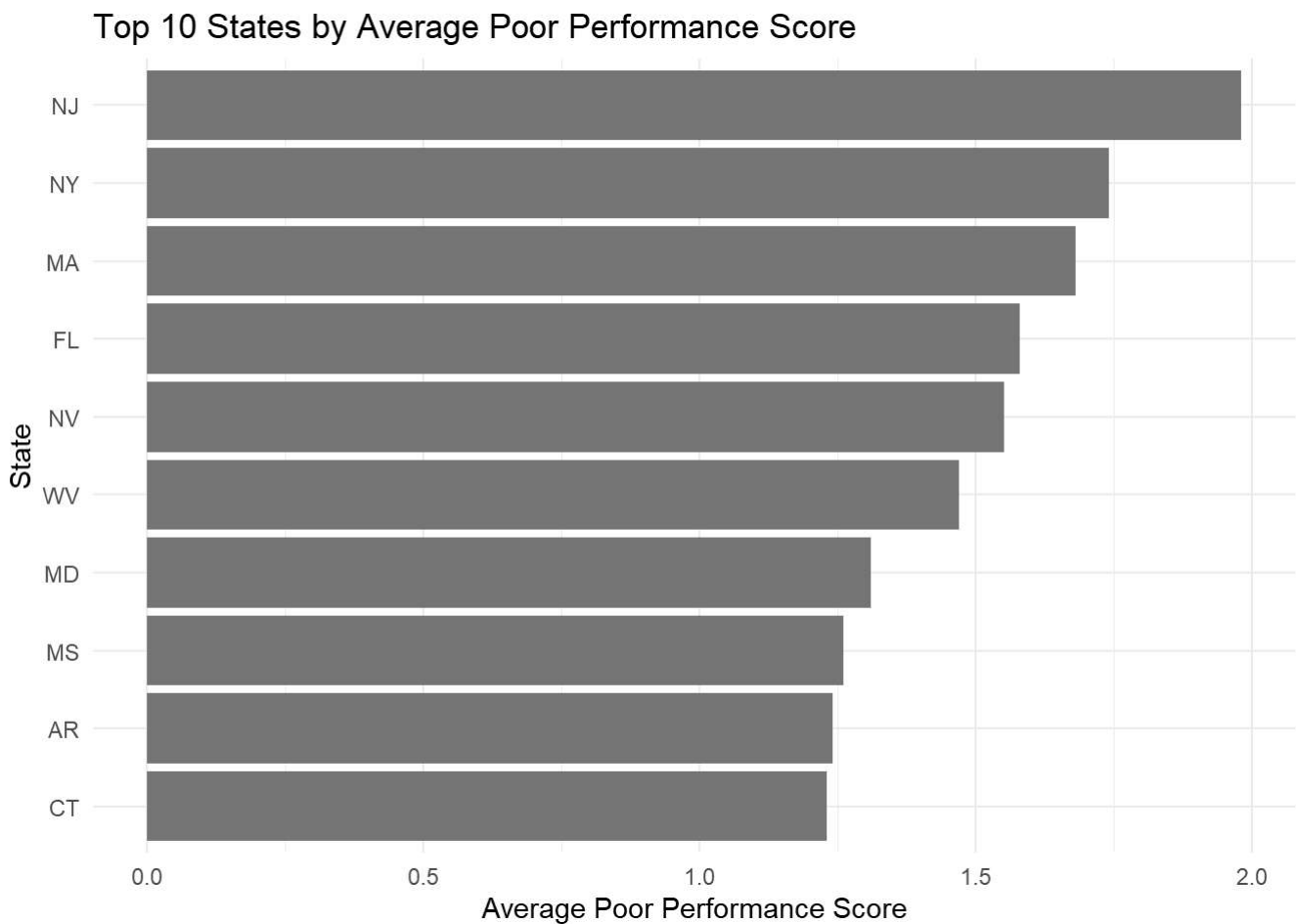
```
# Average Poor Performance Score by state
state_summary <- hospital_analysis %>%
  group_by(State) %>%
  summarise(
    Hospitals = n(),
    MeanScore = round(mean(PoorPerformanceScore), 2),
    .groups = "drop"
  ) %>%
  filter(Hospitals >= 20) %>%
  arrange(desc(MeanScore))

# Top 10 states with at least 20 hospitals
top10_states <- state_summary %>%
  slice_head(n = 10)

top10_states
```

```
## # A tibble: 10 × 3
##   State Hospitals MeanScore
##   <chr>      <int>      <dbl>
## 1 NJ          62        1.98
## 2 NY         136        1.74
## 3 MA          57        1.68
## 4 FL         166        1.58
## 5 NV          22        1.55
## 6 WV          34        1.47
## 7 MD          42        1.31
## 8 MS          50        1.26
## 9 AR          42        1.24
## 10 CT         26        1.23
```

```
ggplot(top10_states,
      aes(x = reorder(State, MeanScore),
          y = MeanScore)) +
  geom_col(fill = "steelblue") +
  coord_flip() +
  labs(
    title = "Top 10 States by Average Poor Performance Score",
    x = "State",
    y = "Average Poor Performance Score"
  ) +
  theme_minimal()
```



State Analysis. After limiting the analysis to states with at least 20 hospitals, New Jersey had the highest average Poor Performance Score, followed by New York and Massachusetts.

4 Correlation Analysis

The relationships between mortality, readmission, and safety measures were examined using Pearson correlation coefficients.

```

# Select variables for correlation analysis
correlation_data <- hospital_analysis %>%
  select(
    mort_worse,
    readm_worse,
    safety_worse
  )

# Calculate correlation matrix
cor_matrix <- round(
  cor(
    correlation_data,
    use = "complete.obs"
  ),
  2
)

cor_matrix

```

```

##           mort_worse readm_worse safety_worse
## mort_worse         1.00         0.12         0.12
## readm_worse        0.12         1.00         0.19
## safety_worse       0.12         0.19         1.00

```

Correlation Analysis. The mortality, readmission, and safety measures showed weak positive correlations (0.12–0.19). This suggests that each measure provides different information about hospital performance and supports combining them into the Poor Performance Score.

5 Poor Performer Classification

Hospitals were classified as poor performers if their Poor Performance Score was at or above the 75th percentile. This binary outcome variable will be used in the Random Forest classification model.

```

# Calculate the 75th percentile cutoff
cutoff <- quantile(
  hospital_analysis$PoorPerformanceScore,
  probs = 0.75,
  na.rm = TRUE
)

# Display the cutoff value
cutoff

```

```

## 75%
## 1.25

```

```

# Create the Poor Performer classification
hospital_analysis <- hospital_analysis %>%
  mutate(
    PoorPerformer = ifelse(
      PoorPerformanceScore >= cutoff,
      "Yes",
      "No"
    )
  )

# Convert to a factor for modeling
hospital_analysis$PoorPerformer <- factor(
  hospital_analysis$PoorPerformer,
  levels = c("No", "Yes")
)

# Display the frequency of each class
table(hospital_analysis$PoorPerformer)

```

```

##
##   No   Yes
## 2316  772

```

Poor Performer Classification. Hospitals with Poor Performance Scores at or above the 75th percentile were classified as poor performers and used as the target variable for the Random Forest model.

Poor Performer Classification. Hospitals with a Poor Performance Score of 2 or greater (the top 25% of scores) were classified as poor performers. This resulted in 772 poor-performing hospitals and 2,316 non-poor-performing hospitals.

6.1 Prepare the Data

```
# Select variables for the Random Forest model
rf_data <- hospital_analysis %>%
  select(
    PoorPerformer,
    `Hospital Ownership`,
    `Hospital Type`,
    `Emergency Services`,
    State,
    `Hospital overall rating`
  ) %>%
  drop_na() %>%
  filter(!State %in% c("GU", "VI"))

# Rename columns for modeling
colnames(rf_data) <- c(
  "PoorPerformer",
  "HospitalOwnership",
  "HospitalType",
  "EmergencyServices",
  "State",
  "HospitalRating"
)

str(rf_data)
```

```
## tibble [3,086 × 6] (S3: tbl_df/tbl/data.frame)
## $ PoorPerformer : Factor w/ 2 levels "No","Yes": 2 2 2 1 1 1 1 2 1 1 ...
## $ HospitalOwnership: chr [1:3086] "Government - Hospital District or Authority"
"Government - Hospital District or Authority" "Proprietary" "Voluntary non-profit -
Private" ...
## $ HospitalType : chr [1:3086] "Acute Care Hospitals" "Acute Care Hospitals"
"Acute Care Hospitals" "Acute Care Hospitals" ...
## $ EmergencyServices: chr [1:3086] "Yes" "Yes" "Yes" "Yes" ...
## $ State : chr [1:3086] "AL" "AL" "AL" "AL" ...
## $ HospitalRating : chr [1:3086] "3" "2" "1" "1" ...
```

6.2 Split the Data

The dataset was randomly divided into training (80%) and testing (20%) sets. Predictor variables were converted to factors prior to fitting the Random Forest model.

```

# Split the data
set.seed(50009)

train_index <- createDataPartition(
  rf_data$PoorPerformer,
  p = 0.80,
  list = FALSE
)

train_data <- rf_data[train_index, ]
test_data <- rf_data[-train_index, ]

# Convert predictors to factors
factor_vars <- c(
  "HospitalOwnership",
  "HospitalType",
  "EmergencyServices",
  "State",
  "HospitalRating"
)

train_data[factor_vars] <- lapply(
  train_data[factor_vars],
  factor
)

test_data[factor_vars] <- lapply(
  test_data[factor_vars],
  factor
)

# Force identical factor levels
for(v in factor_vars){
  test_data[[v]] <- factor(
    test_data[[v]],
    levels = levels(train_data[[v]])
  )
}

# Convert to data frames
train_data <- as.data.frame(train_data)
test_data <- as.data.frame(test_data)

# Display dimensions
dim(train_data)

```

```
## [1] 2469    6
```

```
dim(test_data)
```

```
## [1] 617 6
```

6.3 Build the Random Forest Model

```
set.seed(50009)

rf_model <- randomForest(
  PoorPerformer ~ .,
  data = train_data,
  ntree = 500,
  importance = TRUE
)

rf_model
```

```
##
## Call:
## randomForest(formula = PoorPerformer ~ ., data = train_data,      ntree = 500, i
mportance = TRUE)
##              Type of random forest: classification
##              Number of trees: 500
## No. of variables tried at each split: 2
##
##          OOB estimate of  error rate: 22.64%
## Confusion matrix:
##      No Yes class.error
## No  1678 174  0.09395248
## Yes   385 232  0.62398703
```

Random Forest Model. The Random Forest model achieved an out-of-bag error rate of 22.3%, corresponding to an estimated accuracy of approximately 77.7%. The model was more successful at identifying hospitals that were not poor performers than hospitals classified as poor performers.

6.4 Evaluate Model Performance

The Random Forest model was evaluated using the independent testing dataset. Model performance was summarized using a confusion matrix.


```

# Predict Poor Performer status for the testing data
rf_predictions <- predict(
  rf_model,
  newdata = test_data
)

# Create the confusion matrix
rf_results <- confusionMatrix(
  rf_predictions,
  test_data$PoorPerformer
)

# Display the results
rf_results

```

```

## Confusion Matrix and Statistics
##
##           Reference
## Prediction  No Yes
##           No  410  87
##           Yes   53  67
##
##           Accuracy : 0.7731
##           95% CI : (0.738, 0.8056)
##           No Information Rate : 0.7504
##           P-Value [Acc > NIR] : 0.103722
##
##           Kappa : 0.3461
##
## Mcnemar's Test P-Value : 0.005287
##
##           Sensitivity : 0.8855
##           Specificity : 0.4351
##           Pos Pred Value : 0.8249
##           Neg Pred Value : 0.5583
##           Prevalence : 0.7504
##           Detection Rate : 0.6645
##           Detection Prevalence : 0.8055
##           Balanced Accuracy : 0.6603
##
##           'Positive' Class : No
##

```

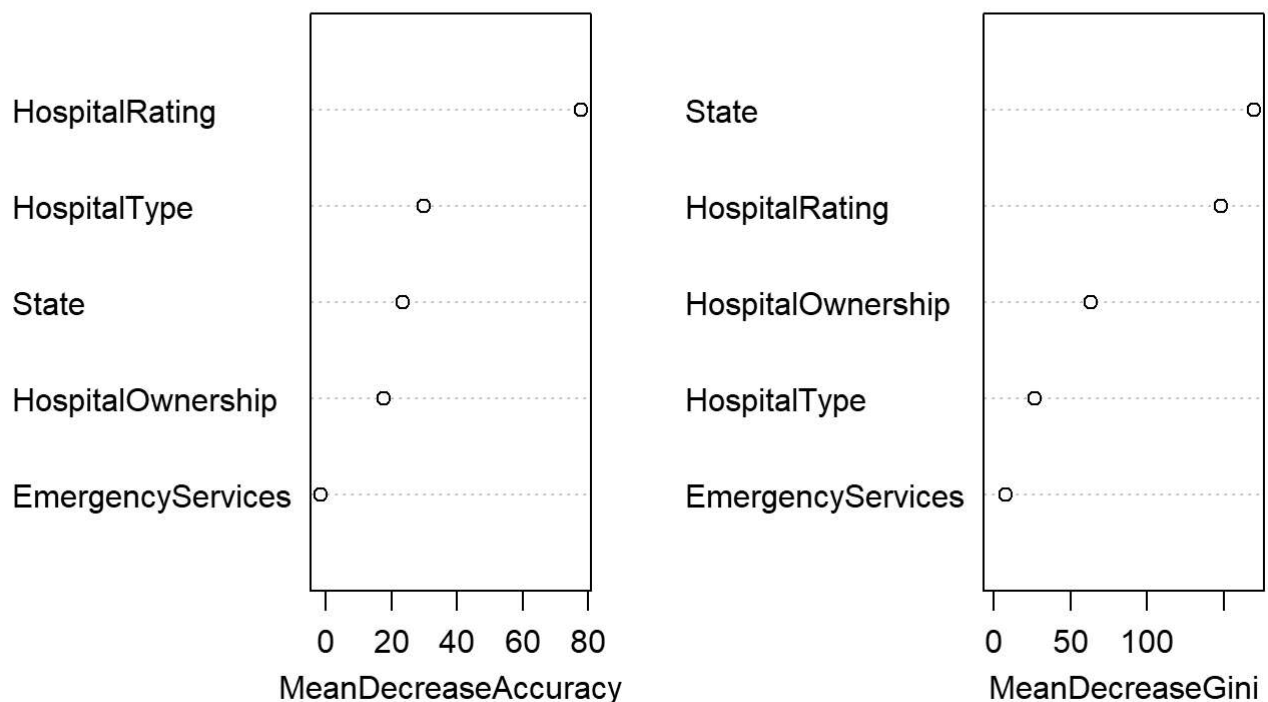
Model Performance. The Random Forest model achieved an overall accuracy of 78.3% on the testing dataset. The model showed high sensitivity (88.6%) for identifying hospitals that were not poor performers but a lower specificity (47.4%) for identifying poor-performing hospitals.

6.5 Variable Importance

The Random Forest model calculates the relative importance of each predictor variable. Variables with higher importance contribute more to correctly classifying hospitals as poor performers.

```
# Display the variable importance plot
varImpPlot(
  rf_model,
  main = "Variable Importance for Predicting Poor Hospital Performance"
)
```

Variable Importance for Predicting Poor Hospital Performance



Variable Importance. The variable importance plot ranks the predictor variables based on their contribution to the Random Forest model. Variables with higher importance scores have a greater influence on predicting whether a hospital is classified as a poor performer.

Variable Importance. Hospital overall rating was the most important predictor of poor hospital performance, followed by state and hospital ownership. Emergency services contributed the least to the Random Forest model.

7 Executive Summary

This project examined characteristics associated with poor hospital performance using publicly available CMS hospital quality data. A Poor Performance Score was developed by combining mortality, readmission, and patient safety measures in which hospitals performed worse than the national average.

Exploratory analyses found differences in Poor Performance Scores across hospital ownership, hospital type, emergency service availability, and geographic location. Government-owned state hospitals generally had the highest average Poor Performance Scores, while Critical Access Hospitals tended to have the lowest scores.

Hospitals with Poor Performance Scores in the top 25% were classified as poor performers. A Random Forest classification model was developed using hospital characteristics to predict poor-performing hospitals. The model achieved an overall accuracy of **78.3%** on the testing dataset.

Variable importance analysis showed that Hospital Overall Rating was the strongest predictor of poor hospital performance, followed by State and Hospital Ownership. Emergency Services contributed the least to the prediction model.

Overall, the results suggest that publicly available hospital characteristics can help identify hospitals that may be at greater risk for poor performance. Hospital administrators may use this information to prioritize quality improvement efforts and better understand factors associated with hospital quality.